MILITARY SPECIFICATION

CARTRIDGE, 25 MM, HIGH EXPLOSIVE INCENDIARY-TRACER

MARK 210 MOD 0

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for the test and acceptance of the percussion primed Cartridge, 25 millimeter (mm), high explosive incendiary-tracer MARK 210 MOD 0, herein referred to as the cartridge.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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MIL-STD-109	Quality Assurance Terms and Definitions.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-644	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition.
MIL-STD-651	Visual Inspection Standards for 20 mm Ammunition and Components.
MIL-STD-1167	Ammunition Data Cards.
MIL-STD-1168	Lot Numbering of Ammunition.
MIL-STD-1235	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1397AS690	Cartridge, 25 mm, HEI-T, MK 210 MOD 0.

PUBLICATIONS

Code of Federal Regulations (CFR)

49 CFR 100-199	Hazardous Materials Regulations, Department of Transportation.

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

REPORTS

U.S. Army Armament Research and Development Command
(Code Ident 19200)

AS12013566	25 mm Ammunition Ballistic Test Methods.

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ASI2013667 Special Ammunition Requirements.

(Copies of standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS.

3.1 General. The cartridge shall comply with Drawing 1397AS690 and other requirements specified herein.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting activity.

3.3 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to a first article inspection.

3.4 Projectile extraction. The force required to extract the projectile from the case shall be not less than 8,900 newtons (N) and not greater than 18,200 N.

3.5 Projectile torque. The projectile shall not rotate in the case when subjected to a minimum torque of 3.4 newton meters (Nm).

3.6 Velocity. The average muzzle velocity of the sample cartridges, conditioned at 18 to 24 degrees Celsius (°C), shall be 1,100 ±25 meters per second (mps). The sample standard deviation shall be not greater than 13 mps.

3.7 Pressure. The average peak chamber pressure of the sample cartridges conditioned at 18° to 24°C, plus 0.72 standard deviation of peak chamber pressure, shall be not greater than 402 megapascals (MPa). The average peak chamber pressure of the cartridges, when functioning at any individual temperature from -54 to 62°C, plus five standard deviations of peak chamber pressure, shall be not greater than 496 MPa. Pressure shall be measured with a piezoelectric type pressure transucer or equivalent.

3.8 Waterproofness. The average velocity (wet) of projectiles of the sample cartridges, conditioned at 18 to 24°C, shall not vary from the average velocity (dry) by more than 30.5 mps. The action time of the (wet) cartridges shall be not greater than 6.0 milliseconds (ms).

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3.9 Accuracy. The standard deviation of impacts in both the horizontal and the vertical directions of the sample cartridge projectiles shall be not greater than 0.77 mils when the sample cartridges are conditioned at 18 to 24°C.

3.10 Action time. The action time of the cartridge shall be not greater than 6.0 ms at any cartridge temperature from -54 to 62°C.

3.11 Projectile function. The projectile shall function upon impact with a target of 0.040-inch nominal thickness aluminum plate 2024 T3 at 0 ±5 degrees obliquity, at a distance of 180 to 220 meters.

3.12 Projectile trace. The projectile shall exhibit a visible trace for a minimum of 3.2 seconds when viewed from the gun muzzle against a dark background.

3.13 Function and casualty. The cartridge shall function at 20 ±10°C without firing defects and casualties.

3.14 Workmanship. All parts and assemblies shall be fabricated, loaded, and assembled in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, scratches, dents, folds, wrinkles, buckles, dirt, grease, oil, rust and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous. Light scratches shall not exceed the defect criteria defined in 6.4.2. All required marking and stamping shall be neat and clearly defined.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein, and the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Terms and definitions. Quality assurance terms and definitions shall be in accordance with MIL-STD-109.

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4.4 First article sample inspection.

4.4.1 First article sample. Prior to regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 914 cartridges, 125 primed cases, and 125 projectiles (see 6.3). The sample shall be manufactured using equipment, processes, and procedures that are similar to those used in regular production. Identification shall be in accordance with MIL-STD-1168.

4.4.1.1 Examination and test. The tests of Table I shall be performed on the first article sample in accordance with the test methods prescribed in 4.6. Except as otherwise specified herein, the tests shall be conducted with samples at 20 ±10°C. Approval shall be based upon eximination of the complete sample for visual and dimensional characteristics listed in 4.5.3 and the tests of Table I. The combining of tests shall be permitted provided that the test results are not affected by this procedure.

TABLE I. First article inspection.

Examination or test	No. of sample units	Acc/rej	Requirement paragraph	Paragraph reference/ inspection method
Projectile, assembly, 25 mm, HEI				
Examination for defects	125	Acc - 0 Rej - 1	3.1	4.5.3.1
Case, cartridge, primed				
Examination for defects	125	Acc - 0 Rej - 1	3.1	4.5.3.2
Cartridge, 25 mm, PGU-22/U (Drawing 1397AS290)				
Examination for defects	125	Acc - 0 Rej - 1	3.1	4.5.3.3
Projectile extraction	32 3/	Acc - 1 Rej - 2	3.4	4.6.1
Propellant contamination	32 3/	Acc - 0 Rej - 1	3.14	4.6.2

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TABLE I. First article inspection.

Examination or test	No. of sample units	Acc/rej	Requirement paragraph	Paragraph reference/ inspection method
Projectile torque	32	Acc - 1 Rej - 2	3.5	4.6.3
Pressure	135 4/	Acc - 0 Rej - 1	3.7	4.6.4
Velocity	45 1/	Acc - 0 Rej - 1	3.6	4.6.4
Action time	135 4/	Acc - 0 Rej - 1	3.10	4.6.6
Waterproofness	50 1/	Acc - 0 Rej - 1	3.8	4.6.4
Accuracy	40	Acc - 0 Rej - 1	3.9	4.6.5
Projectile function	50	Acc - 2 Rej - 3	3.11	4.6.7
Projectile trace		-------	3.12	4.6.8
-32° ±3°C	50	Acc - 4 Rej - 5		
+21° ±3°C	50	Acc - 3 Rej - 4		
+52° ±3°C	50	Acc - 4 Rej - 5		
Function/casualty M242 Gun	150 2/	-------	3.13	4.6.9
Function/casualty AV-8/25mm Gun System	150 2/	-------	3.14	4.6.10

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TABLE I. First article inspection. (Continued).

1/ Ammunition shall be conditioned at 18 to 24°C.

2/ The sample shall be rejected if malfunctions or firing casualties in excess of the "Acc" number on Table III occur.

3/ Quantity combined for tests.

4/ Forty-five cartridges each shall be conditioned at -54° ±3°C, +21° ±3°C, +62° ±3°C.

4.4.1.2 First article sample rejection. Failure of the sample to comply with the requirements of the drawings and specifications specified herein shall result in sample disapproval. Determination as to acceptability of any first article sample shall be based upon results of initial test only and no second tests shall be permitted on that first article. The acceptance/rejection criteria for the first article sample shall be as specified in Table I. All first article units shall be inspected 100 percent for critical defects.

4.5 Quality conformance inspection.

4.5.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.5.1.1 Production lot. The production lot shall be not less than 25,000 cartridges and not greater than 200,000 cartridges. A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with referenced drawings.

b. Like parts and assemblies (cases; primers; projectiles; fuzes; tracer and ignition systems) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.5.1.2 Lot identification. Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall further be identified by a national stock number assigned by the contracting activity.

4.5.2 Quality conformance inspection sampling. A 100-percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Each part and each assembly found to be defective shall be rejected and removed from the lot for disposition.

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4.5.2.1 Sampling and acceptance criteria. The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235 or approved contractor sampling plans. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class (see 4.5.2.1.1c). In addition, where three or less defects appear in a classification or where MIL-STD-1235 is used, sampling plans applicable to a class shall not be used for major defects (see 6.5).

4.5.2.1.1 AQLs. The following AQLs are assigned to major and minor defects:

		AQL (%)
a.	Individuals:	
	Major:	0.25
	Minor:	0.40
b.	Class basis:	
	Major:	1.50
	Minor:	1.50
c.	Individual defects associated with the class basis:	
	Major:	0.40

4.5.2.2 Machine inspection. Inspection of critical, major and minor characteristics shall be performed 100 percent by a qualified and periodically verified inspection probe in lieu of sampling the product utilizing the sampling procedures defined in footnote 2 of 4.5.3.

4.5.3 Classification of defects.[1/] The classification of defects shall be as follows:

4.5.3.1 Projectile assembly, 25 mm HEI-T.

Categories and defects	Method of inspection	AQL (%)
CRITICAL:		
HEI charge depth	Gage	100[2/]

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Categories and defects	Method of inspection	AQL (%)
MAJOR:		
None defined.		
MINOR:		
201. Workmanship (see 3.13)	Visual	1.5

4.5.3.2 Case, cartridge, primed.

Categories and defects	Method of inspection	AQL (%)
CRITICAL:		
1. Ignition system is properly assembled	Visual or gage	100[2/]
2. Primer above flush	Gage	100[2/]
MAJOR:		
101. Depth of primer seating oversize . . .	Gage	0.25
MINOR:		
201. Waterproof sealant missing around primer	Visual	0.40
202. Primer crimp missing	Visual	0.40
203. Workmanship (see 3.13)	Visual	1.5

4.5.3.3 Cartridge (see Drawing 1397AS690).

Categories and defects	Method of inspection	AQL (%)
CRITICAL:		
1. Split or perforated case	Visual	100[2/]
2. Crack or split in projectile	Visual	100[2/]
3. Fuze nose cap missing	Visual	100[2/]
4. Crimp missing (case/projectile) . . .	Visual	100[2/]

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	Categories and defects	Method of inspection	AQL (%)
5.	Low propellant charge	Gage	100[2/]
MAJOR:			
101.	Fuze loose	Gage[3/]	0.25
102.	Overall length, max	Gage	100[2/]

	Categories and defects	Method of inspection	AQL (%)
103.	Improper or incomplete crimp (case/projectile)	Visual	0.25
104.	Profile and alignment, max	Gage	100[2/]
105.	Crack, split or dent in fuze nose cap	Visual	0.25
106.	Crocked, loose, missing or inverted primer	Manual/ primer	0.25
MINOR:			
201.	Marking incorrect, incomplete illegible or missing	Visual	1.5
202.	Scratch, dent, buckle, bulge, wrinkle or fold in case	Visual	0.40
203.	Incorrect type cartridge or components	Visual	0.40
204.	Gap between fuze and projectile body .	Visual	0.40
205.	Workmanship (see 3.13)	Visual	1.5

[1/] MIL-STD-651 shall apply in defining and evaluating cartridge visual defects.

[2/] If machine inspection is not utilized, inspection verification of prior 100-percent inspections shall be performed in accordance with MIL-STD-105, minimum size shall be 315 with zero defects allowed.

[3/] The cartridge shall be classed defective, when the fuze turns with respect to the projectile body upon application of the required torque in a counterclockwise direction.

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4.5.4 Test. The tests of Table II shall be performed on each cartridge lot in accordance with the test methods in 4.6. Unless otherwise specified herein, tests shall be conducted with samples at 20 ±10°C. Sample size and acceptance criteria for each test shall be as specified herein. Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be throughly mixed before being divided into samples for the various tests. The combining of testsis permitted provided that the test results are not affected by this procedure.

4.5.4.1 Firing defects and casualties. Firing defects and associated criteria shall be as specified in Table III. For the defect definitions, see AS12013566.

4.5.4.2 Unlisted firing defects. The lot shall be suspended and referred to the contracing activity for disposition if a malfunction or casualty not covered by this specification occurs in any firing test and indicates that the product is unsuited for the purpose intended.

4.5.5 Packing, packaging and marking inspection. MIL-STD-644 shall be used for the inspection of packaging, packing and marking as applicable to the drawing.

4.6 Test methods and procedures.

4.6.1 Projectile extraction. The method of test shall be as specified in AS12013566.

4.6.2 Propellant contamination. The propellant from each of the cartridges tested for projectile extraction shall be examined visually for contamination as specified in AS12013566.

4.6.3 Projectile torque. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque, to the specified requirement, shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case detectable by misalignment in the scratch mark shall be recorded.

4.6.4 Velocity, pressure, and waterproofness. The methods of test shall be as prescribed in AS12013566. Statistics necessary for testing of results against acceptance criteria shall be computed (see 6.6). For velocity data, a correction factor of 0.46 mps shall be applied to the recorded velocity at the measured range to obtain muzzle velocity. The method of computation shall be:

$$V_{muzzle} = V_{measured} + 0.46\ D$$

where:

D = Distance from muzzle to midpoint between velocity measuring devices.

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TABLE II. Quality conformance inspection.

Test[1/]	Sample size production lot	Requirement paragraph	Test paragraph	Combined		
				Pass	Fail	Retest
Projectile extraction[2/]	20	3.4	4.6.1	1 1	2(min) 4(max)	20
Propellant contamination[3/]	20	3.13	4.6.2	0	1	...
Projectile torque[4/]	20	3.5	4.6.3	0	2	20
Velocity[5/], [13/]	20	3.6	4.6.4			40
Pressure[6/], -54° ±3°C	20	3.7	4.6.4			40
-21° ±3°C	20					40
-62° ±3°C	20					40
Waterproofness[7/], [13/]	20	3.8	4.6.4			40
Accuracy [8/], [13/]	40	3.9	4.6.5			80
Action time[9/], -54° ±3°C	20	3.10	4.6.6	0	1	...
+21° ±3°C	20			0	1	
+62° ±3°C	20			0	1	
Projectile function[10/]	50	3.11	4.6.7	2	5	50
Projectile trace[11/]	50	3.12	4.6.8	3	7	50
-32° ±3°C	32			3	6	32
+21° ±3°C	32			2	5	32
+52° ±3°C	32			3	6	32
Function and casualty[13/] M242 Gun	150	3.13	4.6.9			300
Function and casualty[13/] AV-8/25 mm Gun System	150	3.13	4.6.10			

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TABLE II. Quality conformance inspection. (Continued).

1/ The lot shall be rejected if in any firing test, one or more of the following malfunctions or the casualties of Table III occur:

a. Premature functioning of the projectile (see 6.4.3).

b. Projectile remaining in bore.

c. Metal parts separation, except rotating band separation (see Footnote 4/ of Table III).

d. Accept or reject criteria for firing casualties are shown in Table III.

An additional shot shall be fired except as otherwise specified herein, and in the function and casualty test where the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characateristics being tested.

2/ Projectile extraction.

a. Failure of 2 or more sample cartridges to comply with minimum requirements shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined samples fail to comply with the minimum requirement.

b. Failure of four or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than one, but less than four cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if five or more cartridges of the combined samples fail to comply with the maximum requirement.

3/ Propellant contamination. The presence of foreign matter in the propellant of any of the test cartridges shall cause rejection of the lot (see 3.14).

4/ Projectile torque. Failure of two or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if two or more cartridges of the combined sample fail to comply with the applicable requirements.

5/ Velocity. If the sample fails to comply with either or both requirements, a second sample of 40 cartridges shall be tested for the failing parameter(s) (average, standard deviation). The lot shall be rejected if the second sample fails to comply with the applicable requirements (see 4.6.4).

6/ Pressure. If the sample fails to comply with the requirements, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the second sample fails to comply with the applicable requirements.

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TABLE II. Quality conformance inspection. (Continued).

7/ Waterproofness. If the average velocity (wet) of the sample fails to comply with the applicable requirement, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 30.5 mps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base of comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirement of 3.6. If any sample cartridge fails to meet the action time requirement, the lot shall be rejected.

8/ Accuracy. If the accuracy of the sample cartridges exceeds the applicable requirement, a second sample of 80 cartridges shall be tested. The lot shall be rejected if the accuracy exceeds the applicable requirements.

9/ Action time. If any sample cartridge fails to meet requirements for action time, the lot shall be rejected.

10/ Projectile function. Failure of five or more sample cartridges to comply with the applicable requirement shall cause rejection of the lot. If more than two or less than five fail to comply, a second sample of 50 cartridges shall be tested. The lot shall be rejected if seven of more cartridges of the combined samples fail to comply with the applicable requirements.

11/ Projectile trace.

a. Ambient. Failure of five or more sample cartridges to comply with applicable requirements shall cause rejection of the lot. If more than two but less than five fail to comply, a second sample of 32 cartridges shall be tested. The lot shall be rejected if seven or more cartridges of the combined sample fail to comply with the minimum requirement.

b. Hot and cold. Failure of six or more sample cartridges to comply with applicable requirements shall cause rejecteion of the lot. If more than three, but less than six fail to comply, a second sample of 32 cartridges shall be tested. The lot shall be rejected if eight or more of the combined sample fail to comply with the minimum requirements.

12/ Function and casualty. The lot shall be rejected if any malfunction or firing casualty of Table III occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number but less than the applicable "Rej" number for either gun system (see 4.6.9 and 4.6.10), a second sample of 300 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number for either gun system.

13/ Ammunition shall be conditioned at 18 to 24°C.

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TABLE III. Firing defects.

	Acc	Rej
Cartridge:		
Misfire	0	2
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
Premature projectile function	0	1 (see 6.4.3)
Primer:		
Primer leak	6	17
Primer perforation	1	4
Loose primer	0	1
Blown primer	0	1
Case:		
Longitudinal split[1/]		
H or S	7	22
G or J	2	7
K, L or M	0	1
Circumferential rupture, partial[1/]		
S, J or K	2	7
G or L	1	5
Circumferential rupture (complete)[1/]	0	1
Detached metal[2/]	0	1
Projectile:		
Rotating band separation[3/]		
Complete or partial separation	0	1
Metal parts separation[4/]	0	1

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TABLE III. Firing defects. (Continued.)

1/ See Figure 1 for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table III for the areas involved shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classified as a complete circumferential rupture.

2/ Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears, shall be classified as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

3/ For classification as a defect, there shall be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing.

4/ Separation or breakup of projectile part(s), as evidenced by recovery of the part(s) or fragment(s) or hole(s) in the fragmentation screen(s), shall be classified as a defect. The lot shall not be penalized for evidence of any individual fragment with a weight not greater than 0.10 grams.

4.6.5 Accuracy. The method of test and measurement of targets shall be as prescribed in AS12013566. The specified range shall be shortened to a minimum range of 100 meters. Statisics necessary for testing of results against acceptance criteria shall be computed (see 6.6).

4.6.6 Action time. The method of test shall be as specified in AS12013566.

4.6.7 Projectile function. The method of test shall be as specified in AS12013566. The aluminum target shall be supported on the outer edges and located as required from the gun muzzle normal to the line of fire. Observation of projectile functioning shall be made from a suitable position (see 3.11).

4.6.8 Projectile trace. The method of test shall be as specified in AS12013566. Visual observation(s) or photographic witness of trace characteristics shall be made from suitable position(s).

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4.6.9 Function and casualty M242 or XM241 gun. The method of test shall be as prescribed by AS12013566. The test sample shall be assembled in belts, then fired in bursts of approximately 25 cartridges at a rate of 200 ±50 rounds per minute. The weapon barrel shall be at ambient temperature at beginning of test and cooled to ambient temperature after each 100 rounds. The test data shall be examined to verify that the requirement of 3.12 has been met.

4.6.10 Function and casualty AV-8/25mm Gun System. The test shall be conducted using the witness panel test setup specified in AS12013566 and the AV-8/25 mm Gun System mounted on a test stand approved by the contracting activity. The gun system shall be loaded with a compliment of 150 cartridges and fired in four 1-second bursts with a 5-minute cooling period between each burst. The fourth burst shall be a fire-out. The firing sequence shall be accomplished at a firing rate of 3,600 +400, -200 rounds per minute. The test data shall be examined to verify that the requirements of 3.12 have been met.

4.6.11 Defective weapon. If any test cartridge firing defect is found to have been caused by a defective test weapon, the firing defect shsll not be valid for lot penalty. In such cases, the defective test weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. If such firing defect is not found to have been caused by the defective weapon, it shall be valid for lot penalty.

5. PACKAGING.

5.1 Preservation, packaging and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the cartridge shall be preserved and packed in accordance with the requirements of the Transportation Packaging Order (TPO) Drawing established by the Government. Packaging and packing shall conform to all the applicable requirements for the shipment of hazardous items as defined in 49 CFR 100-199

5.2 Marking. Unless otherwise specified in the contract or purchase order (see 6.2.1), marking for shipment shall be in accordance with MIL-STD-129. In addition, exterior containers shall show the caution markings required by 49 CFR 100-199, indicating the appropriate classification of ammunition.

5.3 Documentation with shipment. When specified in the contract or purchase order (see 6.2.2), each shipment of ammunition shall contain data cards in accordance with MIL-STD-1167, lot number assignment in accordance with such items as case, primer, ignition system, propellants, charge weights, projectile, and lot identification.

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6. NOTES.

6.1 Intended use. The cartridges are intended for use in 25 mm automatic weapons.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Place of inspection, if not at place of manufacture (see 4.1).

c. Disposition of rejected samples (see 4.5.2).

d. Detailed preservation, packaging, packing and marking instructions (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph	Data requirements	Applicable DID	Option
5.3	Ammunition Data Cards	DI-E-2001	---
3.2	Description of Manufacture	DI-P-1614	---

(DIDs related to this specification, and identified in Section 6, will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition function should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article inspection sample. The contracting activity may waive the requirement for a first article inspection sample if the contractor has recently demonstrated his ability to produce this item.

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6.4 Definitions.

6.4.1 Projectile coating defect criteria. Projectiles indicating exposure of bare metal or underlying phosphate or organic pretreatment shall be considered unsatisfactory if any of the following conditions exist:

a. Any area exceeding 0.090 square inches.

b. More than one area exceeding 0.045 square inches.

c. More than five areas of 0.021 square inches or less.

6.4.2 Premature functioning. Is defined as the occurrence of any of the following:

a. Any function within the weapon.

b. Any function within 1 meter of the weapon.

c. Any functional within 10 meters of uninterrupted flight from the weapon.

6.5 AQLs. The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency adminstrative burden which might result from an exclusive assignment of individual defect AQLs. The option also permits flexibility where continuous or sequential sampling inspections utilizing MIL-STD-1235 (Ord), Contractor Plan NB 502-1, or Contractors Sequential Ration Plan for acceptance is integrated into the manufacturing process.

6.6 Computations. Standard deviation. Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$s = \frac{\sum (X_i - \overline{X})^2}{(n - 1)} \quad \text{or equivalent}$$

Where: X_i = each individual value

$\overline{X}$ = sample arithmetic mean

n = sample size

Preparing activity:
NAVY (AS)

Project (1305-N005)

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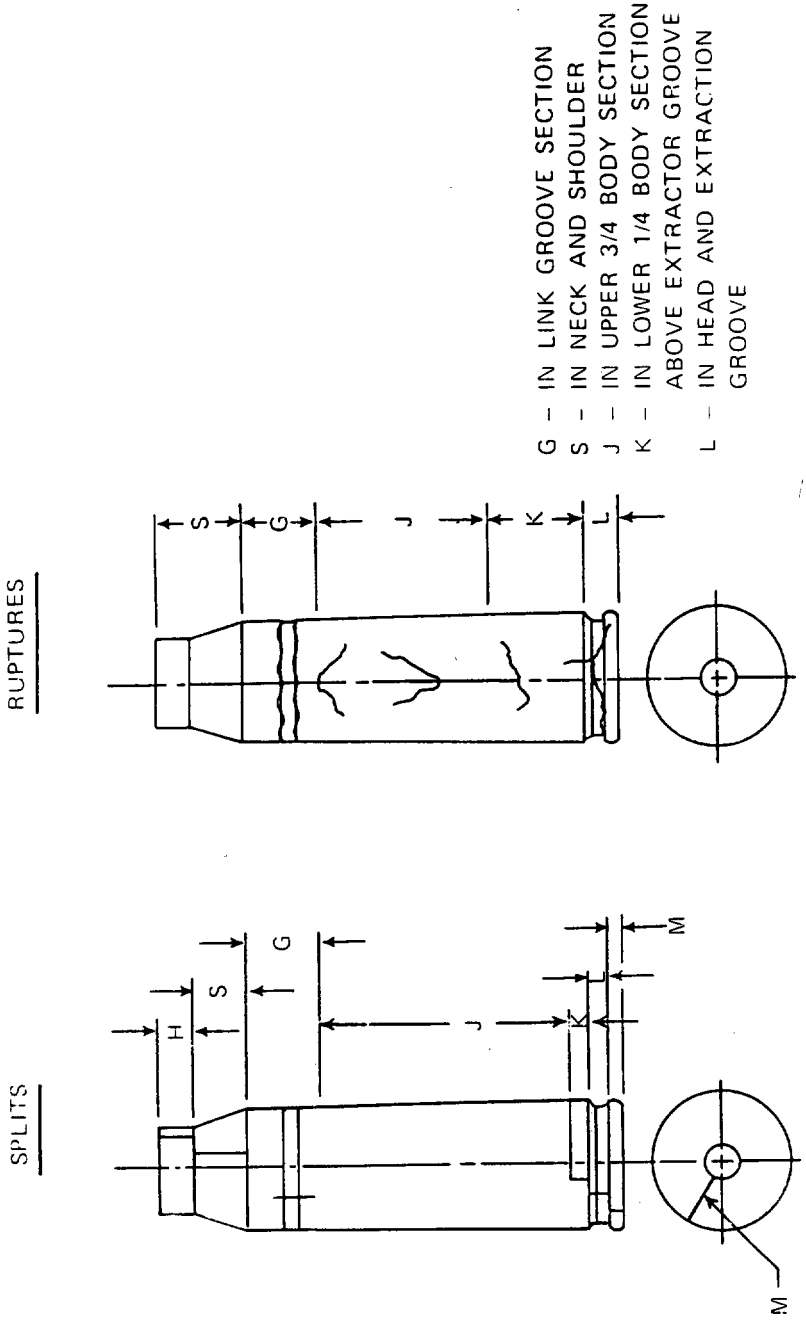


FIGURE 1. Classification of splits and ruptures.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85655 (AS)	CARTRIDGE, 25 MM, HIGH EXPLOSIVE INCENDIARY-TRACER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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